

Software Development Life Cycle (SDLC)

System Requirements Specification Draft

1. System Overview

The **Stitch Hotel Booking System** is a web-based application designed to allow users to browse hotel information, select rooms, and securely complete booking payments.

2. Functional Requirements

- **Authentication:** Entry point with a secure user login page.
- **Navigation:** Core navigation bar including Home, Amenities, Suites/Rooms, About Us, Location, and Logout.
- **Booking Flow:** Interface for users to select specific rooms and proceed to checkout.
- **Payment:** Secure payment checkout integration post-room selection.

3. Technical Stack

- **Frontend Framework:** React (Developed in Visual Studio Code)
- **Database:** Neon DB (Serverless PostgreSQL)
- **Payment Gateway:** Stripe

4. Pending SDLC Phases (To Be Expanded)

4.1 System Design

Database schema mappings, Entity-Relationship diagrams, UI/UX wireframes, and API routing architecture.

4.2 Implementation Details

Backend logic architecture and frontend state management definitions.

4.3 Testing Strategy

Outlines for unit testing, integration testing (especially for Stripe webhooks), and User Acceptance Testing.

4.4 Deployment Plan

Cloud hosting environment and CI/CD workflow specifics.